

Deployment challenges addressed: chemistry mismatch • manufacturing variability • cross-condition shift

Target cell
early-life
cycling data

Discharge
voltage
curves $Q(V)$

$\Delta Q(V)$
descriptor

Logistic
screen

Early defect screen

Sec 8
AUC 0.78
50-cycle lead

SOH curve
(first 30
cycles)

CNN
fingerprint
encoder

LSTM
fine-tune +
MC-Dropout

One-step SOH + 95% CI

Sec 7
RMSE 0.014 fine-tuned
53.5% over scratch

MIT LFP pretraining
(chemistry-aligned)

Direct
multi-horizon
decoder

RUL + conformal PI

Sec 9
full coverage
49-cycle median error

LFP foundation corpus
155 cells (MIT + HUST + SNL)

